

Name: _____

Period: _____

Solving By Substitution Practice

$$1. \begin{cases} r+s+t=10 \checkmark \\ r-s-t=-2 \\ 2r+t=12 \end{cases}$$

$$\begin{array}{r} r+s+t=10 \\ -r-s \quad -r-s \\ \hline t=10-r-s \end{array}$$

$$\begin{array}{r} r-s-(10-r-s)=-2 \\ r-s-10+r+s=-2 \end{array}$$

$$\begin{array}{r} 2r-10=-2 \\ +10 \quad +10 \\ \hline \end{array}$$

$$\begin{array}{r} 2r=8 \\ \hline 2 \quad 2 \end{array}$$

$$r=4$$

$$2(4)+t=12$$

$$\begin{array}{r} 8+t=12 \\ -8 \quad -8 \\ \hline \end{array}$$

$$t=4$$

$$(4)+s+(4)=10$$

$$\begin{array}{r} s+8=10 \\ -8 \quad -8 \\ \hline \end{array}$$

$$s=2$$

(4, 2, 4)

$$2. \begin{cases} a-3b+2c=5 \checkmark \\ 2a-6b+4c=10 \\ a+2b=15 \end{cases}$$

$$\begin{array}{r} a-3b+2c=5 \\ +3b-2c \quad +3b-2c \\ \hline a=5+3b-2c \end{array}$$

$$2(5+3b-2c)-6b+4c=10$$

$$\begin{array}{r} 10+6b-4c-6b+4c=10 \\ \hline \hline \end{array}$$

$$10=10$$

IMS

$$3. \begin{cases} h-4j+3k=16 \\ 2h-8j+6k=8 \\ h-2j+k=5 \end{cases}$$

$$\begin{array}{r} h-4j+3k=16 \\ +4j-3k \quad +4j-3k \\ \hline h=16+4j-3k \end{array}$$

$$2(16+4j-3k)-8j+6k=8$$

$$\begin{array}{r} 32+8j-6k-8j+6k=8 \\ \hline \hline \end{array}$$

$$32 \neq 8$$

No Solution